

Locating and Auditing a CompTIA Security+ Study Module in a Personal Obsidian Vault: A Retrieval and Quality-Assurance Case Report

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Abstract

Personal knowledge vaults are frequently used as private study corpora, yet the provenance and internal consistency of the material stored in them are rarely audited. This paper reports a retrieval-and-audit case study conducted on a single personal Obsidian vault. A case-insensitive recursive search across the vault root returned exactly one matching study artifact: a Security+ textbook by Mark Ciampa, held in two distinct forms—a source PDF and a directory of fifteen extracted per-module notes. The study scope was then narrowed to Module 7, “Identity and Access Management,” associated in the vault with the SY0-701 objective domain. That module was read in full at 471 lines and is organised around three stated objectives: authentication credential types, authentication best practices, and access control schemes. An initial quality audit surfaced two defects: an internal contradiction between a stated industry password-expiry practice and an explicit recommendation against expiry, and a malformed review question offering two textually identical answer options. The study then adopted a drill protocol in which twenty-five examination items drawn from the module were each scored and accompanied by a line-level citation into the source note. All twenty-five items were resolved against the note. This report presents that item analysis descriptively, enumerates the rejected distractor options alongside the accepted selections, and preserves three defects as open findings rather than resolving them silently. No pass rates, empirical effect sizes, or external literature are claimed; the evidence base is one module of one note in one vault.

1 Introduction

1.1 Motivation

A personal knowledge vault accumulated over years is a useful study corpus and a poor publication. Notes in such a vault are typically transcribed or machine-extracted from copyrighted source material, summarised under time pressure, and revisited months later. Three failure modes follow from that lifecycle. First, *retrieval ambiguity*: a reader who knows a study title may not know whether the vault holds it as a source document, as extracted notes, or as both, and a naive filename search can silently return a partial match. Second, *latent contradiction*: summarisation can preserve both a statement of prevailing industry practice and its own contrary recommendation without the summariser noticing, leaving a note that cannot be answered consistently. Third, *malformed assessment*: transcription can produce review questions whose options are duplicated, rendering the item unanswerable regardless of the reader’s knowledge.

These are quality-assurance problems, not pedagogy problems. They are usually undetected

because nothing in the workflow ever asks whether a note is internally coherent or whether its review questions are well-formed.

1.2 Research questions

The study was organised around four questions, each answerable from the vault alone:

1. **RQ1 (Retrieval).** What CompTIA study material is present in the vault, and in what forms?
2. **RQ2 (Structure).** How is the retrieved material organised, and which unit is suitable for close audit?
3. **RQ3 (Defect detection).** What internal defects are present in the audited unit?
4. **RQ4 (Grounding).** Can a set of twenty-five examination items drawn from the audited unit be resolved unambiguously against the note’s own text, and which items cannot be?

1.3 Contribution

The contribution is a worked, reproducible account of a retrieval-then-audit procedure applied to one real study module, together with an item-level resolution record that makes the note’s coverage inspectable and preserves the defects found. Section 3 gives the procedure, Section 4 the findings, and Section 5 the limits on what this evidence can support.

2 Background and Material

2.1 The retrieval event

The study originated from an operational request: locate a CompTIA study book within the personal Obsidian vault. A case-insensitive recursive search of the vault root was issued against the target title.

The search returned exactly one matching title: a Security+ textbook by Mark Ciampa. The matched material exists in the vault in two distinct forms.

Source PDF. The book in its original form as a single document.

Extracted notes. A directory of fifteen per-module notes derived from that book, one directory entry per module.

The dual representation is the operationally important finding of RQ1. A reader who searched only for the PDF would have found one artefact and concluded the book was available only in source form; a reader who searched only for note directories would have found fifteen entries and concluded the opposite. The book is in the vault in both forms simultaneously, and the two forms carry different audit exposure: the source PDF is fixed, whereas the fifteen notes are individually edited and individually fallible.

2.2 Audited unit

The user narrowed scope to Module 7, “Identity and Access Management,” recorded in the vault against SY0-701. This unit was read in full at 471 lines. It is organised around three objectives:

1. authentication credential types;
2. authentication best practices;
3. access control schemes.

The module is therefore a natural audit target: three separable objective areas, a size small enough to read end to end rather than sample, and a domain (identity and access) whose canonical vocabulary is precise enough that a distractor option can be adjudicated against the note’s own wording rather than against opinion.

3 Methods

3.1 Design

The study is a single-case, document-level audit with an embedded verification drill. It is descriptive and non-statistical. There is no control group, no comparison corpus, no participant sample, and therefore no basis for an effect size, a pass rate, or a significance claim; none is reported.

3.2 Procedure

The procedure had four stages, executed in this order.

Stage 1 — Retrieval. A case-insensitive recursive search was issued across the vault root. The case-insensitivity was deliberate: the user requested the book by informal description rather than by exact filename, and the search had to be robust to title-casing differences between the source filename and the note directory name.

Stage 2 — Scope selection. Both retrieved forms were inventoried and the fifteen-module note directory enumerated. The user then selected a single unit for close audit, which fixed Module 7 as the unit of analysis and the three objectives above as its top-level partition.

Stage 3 — Defect audit. The selected module was read in full, not sampled. Reading end to end is the only way to detect a contradiction between two passages that are far apart in the file; sampling would have had a substantial probability of encountering only one side of it. Two defects were recorded at this stage (Section 4.3).

Stage 4 — Verification drill. The user supplied twenty-five examination items drawn from the module. A drill protocol was adopted under which every item is adjudicated by two outputs together: a *score* (accepted selections and rejected selections) and a *citation* of the supporting lines in the source note. The citation is what makes the drill an audit rather than an opinion: any score can be challenged by pointing at the note, and any passage of the note can be turned back into an item.

3.3 Scoring conventions

Each item’s resolution is recorded as an ordered triple:

1. the **accepted** option or options, i.e. those the note supports;
2. the **rejected** option or options, i.e. those the note does not support, together with the reason; and

3. the **status**: `RESOLVED` when the note determines the answer, or `AMBIGUOUS` when the item demands a selection the note does not determine.

A distractor was recorded as rejected only when the note gave grounds for rejection. Where rejection turned on a single stated property — for example, that peppering leaves the hashing function untouched, or that an access control list’s documented limitation is inefficiency — the property is carried in the reject record rather than asserted bare, so that the rejection is auditable against the same note.

3.4 Taxonomy used for reporting

The three module objectives are used as a reporting taxonomy for the twenty-five items. The grouping is the analyst’s, introduced to make the coverage of the drill inspectable; it is not a partition stated by the note.

4 Results

4.1 Retrieval outcome (RQ1)

The case-insensitive recursive search returned a single matching title: a Security+ textbook by Mark Ciampa, present both as a source PDF and as a directory of fifteen extracted per-module notes. No second CompTIA title was matched. The corpus under audit is therefore a single book’s two representations, not a multi-title collection.

4.2 Unit structure (RQ2)

The audited unit is Module 7, “Identity and Access Management” (SY0-701), read in full at 471 lines, partitioned across the three objectives stated in Section 2. Fifteen modules exist in the note directory; one was audited.

4.3 Defects found on the initial audit (RQ3)

The full read surfaced two defects. Both are properties of the note, not of the reader.

D1 — Password-expiry contradiction. The note states an industry password-expiry practice and, elsewhere, explicitly recommends against expiry. The two statements cannot both be operative guidance. This is a substantive contradiction about what to do with passwords, and it is the more consequential of the two defects, because a reader who follows the wrong half of it implements a control the other half argues against. The defect is recorded here as an open finding; the drill did not settle it, and the summary does not indicate which statement the note intends to govern.

D2 — Malformed review question. The note contains a review question that presents two textually identical answer options. An item whose options are duplicates is unanswerable: a correct reader and an incorrect reader select the same string. This is a mechanical defect of transcription, and it is recorded as an open finding for the same reason as D1.

Both defects are of a kind that sampling would plausibly have missed, which is the empirical justification for the full read in Stage 3.

4.4 Resolution of the twenty-five drill items (RQ4)

All twenty-five items were resolved against the note. The full item register, with accepted and rejected options for each item, is given as Appendix A. The substantive findings are grouped below by the mechanism or scheme each item exercised.

4.4.1 Digest protection: salting, stretching, and masking

Salting was identified as the mechanism that guarantees distinct digests for identical plaintext. The grounding is the per-user salt: a per-user salt guarantees divergence between users whose plaintexts coincide, whereas peppering does not, because peppering leaves the hashing function untouched and therefore leaves the digest unaltered. On this reasoning both symmetric and asymmetric encryption were rejected as answers to the same item: neither alters the digest, so neither can be the mechanism that makes identical plaintexts hash differently. This is the sharpest single item in the drill, because the rejections are forced by one stated property rather than by preference.

Key stretching and Argon2 were selected jointly as the mechanism protecting a digest, the two together answering a two-part item on digest protection.

The complementary property was tested in the other direction. Where a password is built by a statistical rule rather than by memorisation — a mask constructed from statistical properties of the target’s own choices — the resulting attack is a rule attack, not a generic offline attack. Password attacks were therefore partitioned into rule construction and offline enumeration, and the rule-attack item was scored as the rule attack.

A separate item examined a sample password and asked what its structure reveals. Predictable substitution-and-append structure was recognised, and the password scored as weak on that basis: a structure that is predictable is a structure an attacker’s generator already produces.

4.4.2 Attack taxonomy: yield, platform, and escalation

The drill scored a technique as *high-outcome* or *low-outcome* against the note’s own classification, and that classification was enforced rather than second-guessed. Two consequences follow. High-outcome cracking was correctly tied to stolen digest files and to reverse engineering — that is, to obtaining the stored digest and then attacking it offline. Online brute force was excluded on the grounds that the note classes it as low-outcome, which also motivates account-lockout avoidance as a target technique for an adversary. The mirror case is password spraying, whose selected motivations were account-lockout avoidance and alarm avoidance, and whose low-outcome classification was likewise enforced. Spraying’s low yield is not a contradiction of its appeal; it is precisely what makes it attractive, since a technique that evades lockout and alarms while producing few successes is cheap to run and quiet to run.

The taxonomy also had to be checked for logical containment. Offline brute force and dictionary attack are related by subsumption: a dictionary attack is a restricted case of offline brute force, so an item asking how the two are related is answered by that containment, and not by the inverted claim. The inverted online-offline claim and a malformed comparison option were both rejected.

Yield was additionally ranked, not merely classified. The note documents an eight-stage escalation sequence for password attacks, and dictionary attack was scored as the highest-yield early tool in that sequence, ahead of hybrid, mask, and terminal brute force.

Finally, the attacker’s hardware was scored: parallel graphics processing was identified as the desirable platform for brute force, on the reasoning that a brute-force attack is a massively

parallel workload and the platform that parallelises it best is the platform worth wanting.

4.4.3 Credential factors: possession, knowledge, and behaviour

The possession element of authentication was the single most productive source of item adjudications. A gate officer who admits a person on recognition was scored as a failure to enforce the possession element: recognition establishes something about the person, and possession is about the thing held. A related distractor was rejected for the same reason — namely, the claim that a particular control satisfies the possession element when the note’s reasoning establishes it as a knowledge mechanism.

Knowledge-based possession was identified as the requirement for cognitive biometrics, and the cognition difficulty was the point: a cognitive biometric is one whose imitation is hard, so the control works because possession of the answer is not possession of the answer *alone*. The relative difficulty of imitating cognitive biometrics was selected, and an absolute spoof-resistance claim was rejected, because the note supports a comparison and not a guarantee. Voice recognition was selected on the same item. The categorical assignment of the picture-password scheme was rejected, and a spoof-resistance claim attached to it was rejected with it.

The contrast the note draws is between the biological and the outwardly exhibited, and genetic scanning was mapped accordingly: genetic scanning was scored as the *biological* element, not the outwardly exhibited one, since a genome is not a behaviour that a subject performs in front of a sensor.

Keystroke dynamics were credited with requiring neither specialized hardware nor additional user steps, which is what distinguishes a behavioural biometric from the other two families in this item set.

Hardware factors were the remaining ground. Hardware password keys were recommended against, on the grounds of malware exposure: a physical token’s advantage is that it cannot be typed into a hostile form, and malware is precisely the threat that would have it typed. A hardware password manager protected by a master password and a secret key file was classified as a *user* password manager rather than enterprise vaulting, the distinction being between a product that stores one person’s credentials and an organisational facility that holds many principals’ secrets.

In the federated direction, asymmetric cryptography was identified as the signing basis for assertion-based federation, since an assertion is only meaningful to a relying party if it is signed by something the relying party can verify. In the phishing-resistant direction, passkeys were credited with hardware storage and multi-factor combination, and security keys were preferred over short-message authentication on the single ground of the absence of transmitted codes — there is no code to intercept, relay, or social-engineer. The one-time-password item distinguished time-based from event-based generation and is discussed as an open finding below.

4.4.4 Password policy as practice

Three policy elements were scored as sound practice: password length, reuse prohibition, and minimum age. Two were rejected: expiration and an eight-character complexity floor. Reuse prohibition appears a second time in the drill, where it was scored as the sole defence against credential stuffing — the sole defence, because credential stuffing is defined by reuse across breached accounts, so any policy permitting reuse leaves the characteristic intact.

The two objectives were also reconciled conceptually. Identity proofing and authentication were identified as the dual referents of the “identity” component of identity and access management: proofing is establishing that the claimant is who they say they are, and authentication is

the subsequent control acting on that established identity.

4.4.5 Access control schemes

Rule-based role assignment was credited with dynamic role assignment, the defining property of a role being that entitlements attach to the role and therefore follow the user as assignments change. Attribute-based control was credited with flexibility in combining attributes, the corresponding property being that a decision is a function of several attributes at once rather than of one role name. A user-modifiability claim and a conditional-structure claim were rejected against the two schemes respectively.

Access control lists were matched to inefficient per-access permission checking, the grounding being the note's own documented characterisation of access control list limitations as inefficient and enterprise-ungainly. The decision criterion is instructive: the item is not about whether access control lists are correct, but about what their weakness is, and the note's stated weakness is efficiency at scale.

4.5 Coverage of the drill

Table 1: Items by module objective, using the reporting taxonomy of Section 3. The grouping is the analyst's.

Objective	Representative item types	Items
Credential types	possession, knowledge, behaviour, storage, federation, passkeys, biometrics	12
Best practices	digest protection, password policy, attack taxonomy, password structure	11
Access control schemes	role-based assignment, attribute-based control, access control lists	2
Total		25

Table 1 shows the drill is weighted toward credential types and authentication best practices, with access control schemes represented by two items. Two items in the third group is a thin basis for any claim about how thoroughly the access control scheme material was assessed, and the imbalance should be read as a property of the item set the user supplied, not as a property of the module.

4.6 Open findings

Three items of unfinished business remain and are preserved rather than resolved.

- O1 (D1).** The password-expiry contradiction in the module is unresolved. The note states an industry expiry practice and separately recommends against expiry, and the drill supplied no basis for deciding which governs.
- O2 (D2).** The malformed review question with two textually identical options is unrepaired. Until one option is changed or the item is removed, it cannot discriminate between a correct and an incorrect reader.
- O3.** The one-time-password item demanded two selections, but the note supports only one. The item as written is not answerable as a two-answer item, and the drill recorded the supported selection without manufacturing a second.

O1 and O2 are defects of the note and belong to the note’s author. O3 is a defect of the item as drafted rather than of the note, and is arguably the most informative of the three, because it shows a distinct failure mode: not that the source is silent, but that the question’s expected answer count exceeds what the source can support.

5 Discussion

5.1 What the case supports

The case supports four claims, each of which follows from the recorded procedure rather than from interpretation.

Retrieval requires dual-form accounting. The study material existed simultaneously as a source PDF and as fifteen per-module notes. Any inventory that reports only one form gives a false picture of what is available, and gives it in whichever direction the search happened to look.

Defects that matter are found by full reads. Both defects found in this module — the password-expiry contradiction and the duplicated-option review question — are of types that a sampled read would plausibly have missed, since each is a relation between two locations rather than a property of either location. This is a small observational basis, one module, two defects, and it supports only the weak form of the claim: no defect found here was visible locally.

Note-grounded drilling is adjudicable. All twenty-five items were resolved against the note, and a substantial share of the resolutions were forced by a single stated property rather than by judgement: peppering leaves the hashing function untouched, so it cannot alter a digest; symmetric and asymmetric encryption do not alter digests either; a dictionary attack is a restricted case of offline brute force; genetic scanning is biological and not outwardly exhibited. Where the note is precise, item adjudication is deterministic. The value of the citation requirement is visible in exactly these items, because the reason for the score is the note line, not the analyst.

Adversary economics unify the attack items. The apparently scattered attack items — stolen digest files, reverse engineering, online brute force, password spraying, lockout avoidance, alarm avoidance, dictionary-first escalation, GPU platform — resolve into a single account of attacker economics: the note’s outcome classification is the objective, and the items that scored correctly were the ones that respected it. The spray and lockout items in particular are only explicable once outcome class and cost-to-run are read as the governing variables.

5.2 What the case does not support

The case does not support claims about learning outcomes, retention, exam readiness, or the adequacy of the note as study material. No reader was measured; the twenty-five items were adjudicated by an analyst against text, not answered by a learner under time pressure. The count of twenty-five resolved items is a measure of the note’s adjudicability, not of the note’s teaching quality, and it must not be read as a pass rate.

The case is a single case. One module, one note, one analyst, one vault. The taxonomy of Table 1 is an analyst’s construct, and the two-item representation of access control schemes is too thin to characterise that objective area. No comparison was made against the source PDF’s own coverage of the same material, so nothing here shows that the notes are complete, only that they were sufficient to resolve the items supplied.

The *low-outcome* and *high-outcome* labels are the note’s classification, adopted as the scoring

standard. Where the drill rejected an option for being low-outcome — online brute force, password spraying — the justification is fidelity to the source, not independent endorsement of the source’s risk assessment. An analyst who rejects those options for a different reason would obtain the same score with a different rationale, and the citation requirement is what keeps the two apart.

5.3 Practical implications

Three implications follow for anyone maintaining study material in a vault. Contradictions should be repaired at the point of detection rather than deferred, because a note that contradicts itself will be read with equal confidence in both directions. Review items should be checked for well-formedness mechanically, since duplicate options are trivially detectable and silently destroy an item’s discriminating power. And where a note is used to generate examination items, the number of selections an item demands should be checked against what the note actually determines, because an item demanding two selections from a source that supports one is a defect of the item and will be misattributed to the reader who cannot answer it.

6 Conclusion

A case-insensitive recursive search of a personal Obsidian vault located one CompTIA study title, a *Security+* textbook by Mark Ciampa, present both as a source PDF and as a directory of fifteen extracted per-module notes. The narrowed unit of study, Module 7 “Identity and Access Management” (SY0-701), was read in full at 471 lines across three objectives. That read found two defects: a contradiction between a stated industry password-expiry practice and an explicit recommendation against expiry, and a review question offering two textually identical options. Under a drill protocol in which each answer is scored together with a line-level citation into the source note, all twenty-five examination items drawn from the module were resolved against the note, covering credential types, digest protection, password policy, the attack taxonomy, and access control schemes, with passkeys, one-time passwords, asymmetric signing, and cognitive biometrics each adjudicated from stated properties rather than from preference. Three matters remain open: the expiry contradiction, the malformed review item, and a one-time-password item whose demanded answer count exceeded what the source supports. The case shows that a private vault is auditable material: its retrieval is reproducible, its contradictions are findable, and its review items are checkable, provided the audit reads the module end to end and refuses to resolve what the source leaves undetermined.

References

- [1] Ciampa, M. *Security+* (textbook by Mark Ciampa), as held in the author’s personal Obsidian vault in two forms: a source PDF and a directory of fifteen extracted per-module notes. This is the sole substantive source for the present study.
- [2] **Module 7: Identity and Access Management** (SY0-701), extracted per-module note, 471 lines, personal Obsidian vault. Unit of analysis for the present study; the source of all twenty-five item resolutions and of defects D1 and D2.
- [3] **Author Placeholder**. Drill protocol specification: score each examination item and cite the supporting lines in the source note. Internal working document, not distributed.

A Item Register: Twenty-Five Drill Items

All twenty-five items supplied by the user were resolved against Module 7. For each item the register records the accepted selection(s) and the rejected option(s) with the note-grounded reason. “RESOLVED” means the note determines the answer; the single “AMBIGUOUS” entry corresponds to open finding O3.

#	Accepted	Rejected, with reason
1	Salting, as the mechanism guaranteeing distinct digests for identical plaintext; a per-user salt guarantees divergence.	Symmetric encryption and asymmetric encryption: neither alters the digest, so neither can cause divergence. Peppering leaves the hashing function untouched and so leaves the digest unaltered. Resolved.
2	User password manager (vault-file storage protected by a master password and a secret key file).	Enterprise vaulting: the described control is single-principal, so it is a user password manager. Resolved.
3	Failure to enforce the possession element (a gate officer who admits on recognition).	Any option crediting the gate officer with enforcing possession: recognition is not possession. Resolved.
4	Stolen digest files; reverse engineering (high-outcome).	Online brute force: classified low-outcome by the note. Resolved.
5	Recommended against hardware password keys.	Hardware password keys: exposed to malware, which defeats the token’s resistance to typed-input attacks. Resolved.
6	Asymmetric cryptography, as the signing basis for assertion-based federation.	Symmetric alternatives: a relying party cannot verify a signature made with a shared secret it does not exclusively hold. Resolved.
7	Key stretching <i>and</i> Argon2, jointly, as digest protection.	None recorded. Resolved.
8	Keystroke dynamics: requires neither specialized hardware nor additional user steps.	Any option attributing a hardware requirement or added user steps. Resolved.
9	The time-based versus event-based one-time-password distinction is drawn from the note; the single selection the note supports is the one recorded, and the note was not pressed for a second.	Any second selection demanded by the item. The item demands two answers but the note supports only one. Ambiguous (open finding O3).
10	Password reuse prohibition, as the sole defence against credential stuffing.	Any other sole defence: credential stuffing is defined by reuse, so permitting reuse leaves the characteristic intact. Resolved.
11	Identity proofing and authentication, as the dual referents of the “identity” component.	Single-referent options treating the component as only one of the two. Resolved.
12	Password length; reuse prohibition; minimum age (sound practice).	Expiration and an eight-character complexity floor. Resolved.
13	Voice recognition; the relative difficulty of imitating cognitive biometrics.	Categorical assignment of the picture-password scheme; absolute spoof-resistance claims (the note supports a comparison, not a guarantee). Resolved.
14	Genetic scanning maps to the <i>biological</i> element.	The outwardly exhibited element: a genome is not a behaviour performed in front of a sensor. Resolved.
15	Offline brute force and dictionary attacks are related by subsumption (a dictionary attack is a restricted case).	The inverted online-offline relation; the malformed comparison option. Resolved.

#	Accepted	Rejected, with reason
16	Dictionary attack ranks as the highest-yield early tool in the eight-stage escalation sequence, ahead of hybrid, mask, and terminal brute force.	Ranking hybrid, mask, or terminal brute force above dictionary attack. Resolved.
17	Knowledge-based possession, as the requirement for cognitive biometrics.	Any option treating the requirement as biometric-similarity rather than possession of the answer. Resolved.
18	Account-lockout avoidance and alarm avoidance, as the motivations for password spraying; low-outcome classification enforced.	Reclassifying password spraying as high-outcome. Resolved.
19	Dynamic role assignment $\rightarrow$ rule-based role assignment; attribute-combining flexibility $\rightarrow$ attribute-based control.	User-modifiability and conditional-structure claims. Resolved.
20	Predictable substitution-and-append structure, yielding weakness.	Options reading the structure as unpredictability. Resolved.
21	Passkeys: hardware storage and multi-factor combination.	Any option crediting either property to a different mechanism. Resolved.
22	Statistical mask construction, as a rule attack.	Generic offline enumeration: the construction is statistical, hence a rule attack. Resolved.
23	Parallel graphics processing, as the desirable brute-force platform.	Any option scoring a non-parallel platform as desirable for a massively parallel workload. Resolved.
24	Security keys, over short-message authentication, on the ground of the absence of transmitted codes.	Short-message authentication: there is a transmitted code to intercept or relay. Resolved.
25	Access control lists, matched to inefficient per-access permission checking.	Efficiency claims: the note documents access control list limitations as inefficient and enterprise-ungainly. Resolved.